SENTINEL

An intrusion detection system that knows when to ask a human.

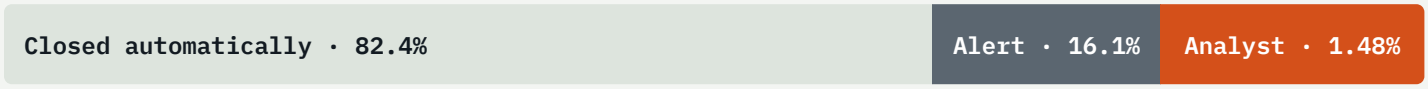
Jai Chavan · Jeet Mane

Guide: **Dr. Kavita Shirsat**

Sentinel handles the traffic it is sure about, asks a security analyst about the rest, learns from verified decisions under fixed rules, and keeps a data warehouse of all its traffic for analysis.

WHAT HAPPENED TO 257,222 UNSEEN TEST FLOWS

held-out CIC-IDS2017 traffic, never used in training



212,015 sure-normal flows closed · **41,401** confident attack alerts · **3,806** doubtful flows for a human

98.9% of the model's mistakes land in the analyst's queue

99.999% correct on the traffic handled without a human

2 of 43,706 attack flows closed as normal

0 false alarms on 213,516 normal flows

HOW IT WORKS

01 A flow arrives — 64 measurements of one conversation. No IPs or ports.

02 CNN-LSTM classifies — Normal, or one of 12 attack types.

03 Three checks — Hesitant? Unfamiliar? Second model disagrees?

04 Route it — Sure: closed or alerted. Any doubt: analyst.

05 Analyst decides — Verdict with reason, beside similar past cases.

Learning loop: verified analyst decisions → candidate model → deployed only if it passes rules set before training; one-click rollback.

DATA AND MODEL

Improved CIC-IDS2017, five days of labelled lab traffic: 2,099,976 flows, **1,714,812** after removing duplicates.



`Conv1D 64 → Pool → Conv1D 128 → LSTM 128 → 13 classes` · + Random Forest (RF) 2nd opinion

ACCURACY · MEAN OF 3 RUNS

Attack vs normal	Paper	RF	CNN-LSTM
Detection rate	96.6%	99.92%	99.75%
Precision	92.1%	99.90%	99.71%
F1 score	94.3%	99.91%	99.73%
Macro-F1 (11 classes)	–	99.23%	98.09%

ATTACKS THE MODEL NEVER SAW IN TRAINING

flows closed as normal: grey model alone → orange with routing

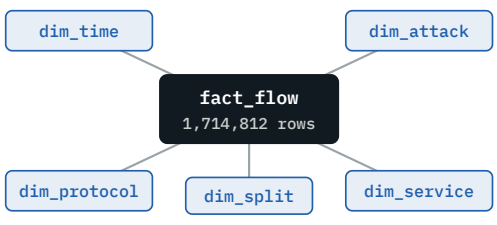
Web attack 97.0% → **5.0%**

Port scan 39.7% → **4.3%**

Botnet 12.4% → **10.2%**

DATA WAREHOUSE AND OLAP

a separate analysis copy; it never feeds the model



ETL loads the dataset and the portal's alerts into a star schema (also snowflake and fact constellation).

25 s full ETL load · **0.3 s** incremental

0.1 ms hourly aggregate vs 2 s on all flows

22 tables: facts, dimensions, marts

roll-up · drill-down · slice · dice · pivot

Python · PyTorch · scikit-learn · FastAPI · SQLite · React

Reference: de Nascimento & Hou, *Safety* 11(4):120, 2025